

Transformations

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Which transformation moves every point of a shape the same distance in the same direction, without turning or resizing it?

.....

2 [2 marks]

Translate the point (3, 2) by the vector $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$

Write down the coordinates of the image.

.....

3 [2 marks]

Reflect the point (3, 5)

(a) in the x -axis

(b) in the y -axis

.....

4 [2 marks]

Rotate the point (4, 0) through 90° anticlockwise about the origin.

.....

5 [3 marks]

Reflect the point (2, 3)

(a) in the line $y = x$

(b) in the line $y = 1$

.....

6

[3 marks]

- (a) Enlarge the point (2, 3) by scale factor 3 about the origin.
(b) A shape is enlarged by scale factor 3. What happens to its area?
-

7

[3 marks]

- Rotate the point (3, 1) about the origin
(a) through 180°
(b) through 90° clockwise
-

8

[3 marks]

- Which of these transformations always give an image congruent to the original shape?
reflection · rotation · translation · enlargement
Explain your answer.
-

9

[4 marks]

- Triangle P has vertices (1, 1), (3, 1) and (1, 4).
Triangle Q has vertices (1, -1), (3, -1) and (1, -4).
Describe fully the single transformation that maps P onto Q.
-

10

[4 marks]

- Triangle T has vertices (1, 1), (2, 1) and (1, 3).
(a) Enlarge T by scale factor 2 about the origin. Write down the new vertices.
(b) Enlarge T by scale factor 2 about the point (1, 1). Write down the new vertices.
-

1 [2 marks]

Rotate the point (2, 5) through 90° clockwise about the origin.

2 [2 marks]

Enlarge the point (4, -2) by scale factor -1 about the origin.

3 [3 marks]

(a) Enlarge the point (5, 3) by scale factor $\frac{1}{2}$ about the origin.

(b) A shape of area 40 cm^2 is enlarged by the same scale factor. Work out the new area.

4 [3 marks]

Triangle A has vertices (0, 0), (2, 0) and (0, 3).

Triangle B has vertices (0, 0), (-2, 0) and (0, -3).

Describe fully the single transformation that maps A onto B.

5 [4 marks]

A shape is reflected in the x-axis, and the image is then reflected in the y-axis.

Show that the result is the same as a single rotation, and describe that rotation fully.

6 [4 marks]

Enlarge the point (4, 3) by scale factor -2 about the centre (1, 1).

7

[4 marks]

- (a) A shape is reflected in the line $x = 3$. Which points of the shape do not move? Explain.
- (b) A shape is rotated about the point $(2, 5)$. Which points do not move? Explain.
-

8

[4 marks]

Describe fully the single transformation equivalent to each pair.

- (a) A translation by $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$ followed by a translation by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$
- (b) A reflection in $y = x$ followed by a reflection in $y = -x$
-

9

[4 marks]

Rotate the point $(5, 2)$ through 90° anticlockwise about the centre $(1, 1)$.

10

[5 marks]

Triangle T has vertices $(1, 1)$, $(4, 1)$ and $(1, 3)$.

- (a) T is enlarged by scale factor -2 about the origin to give T' . Find the vertices of T' .
- (b) Work out the area of T.
- (c) Work out the area of T' .
-